QUANTUM INFORMATION & CYBER-DECEPTION

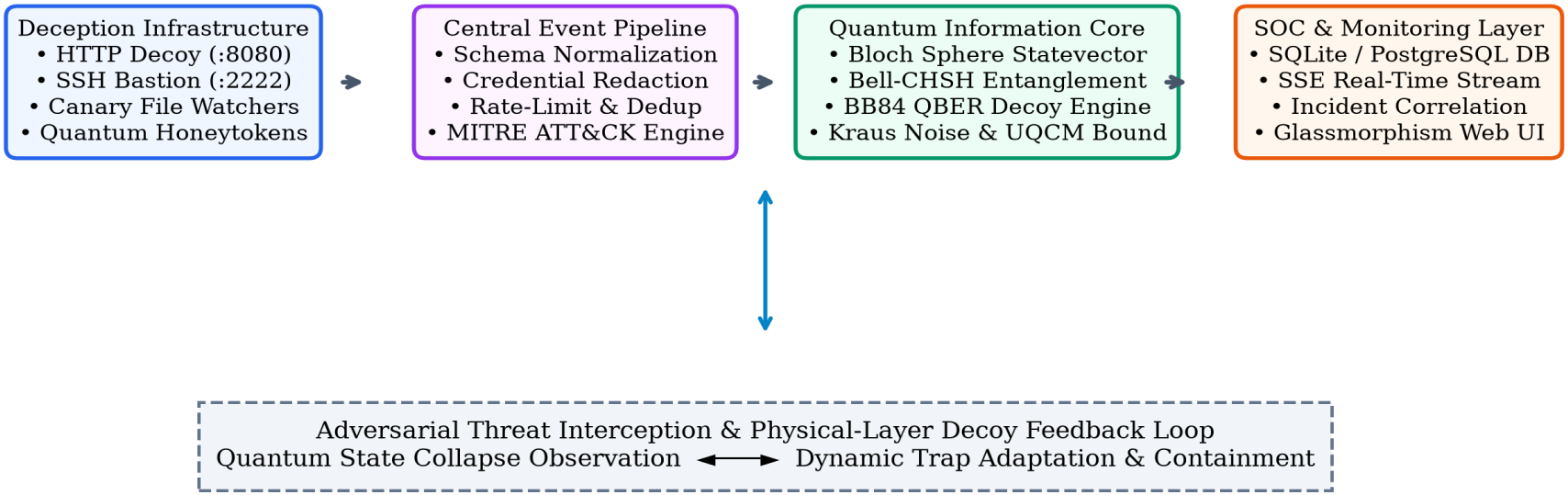
QuantumGuard: Entanglement Verification & State-Collapse Honeypot Architecture

A Hybrid Cyber-Physical Deception Platform for Real-Time Threat Interception and Post-Quantum Network Defense

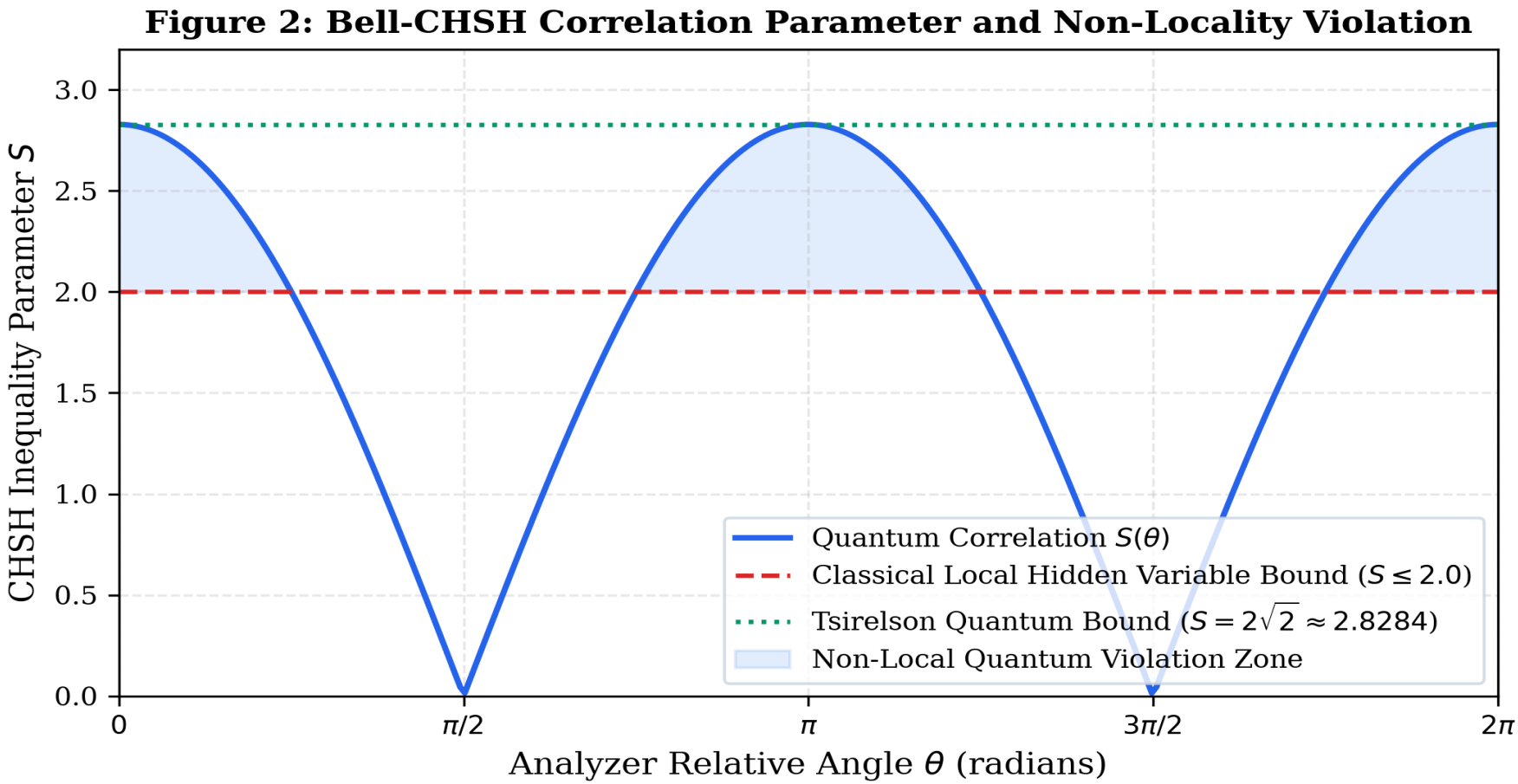
Presented by: Aman Verma & Research Engineering Group | Project Defense 2026

System Architecture & Hybrid Deception Flow

Figure 1: QuantumGuard System Architecture & Hybrid Deception Pipeline

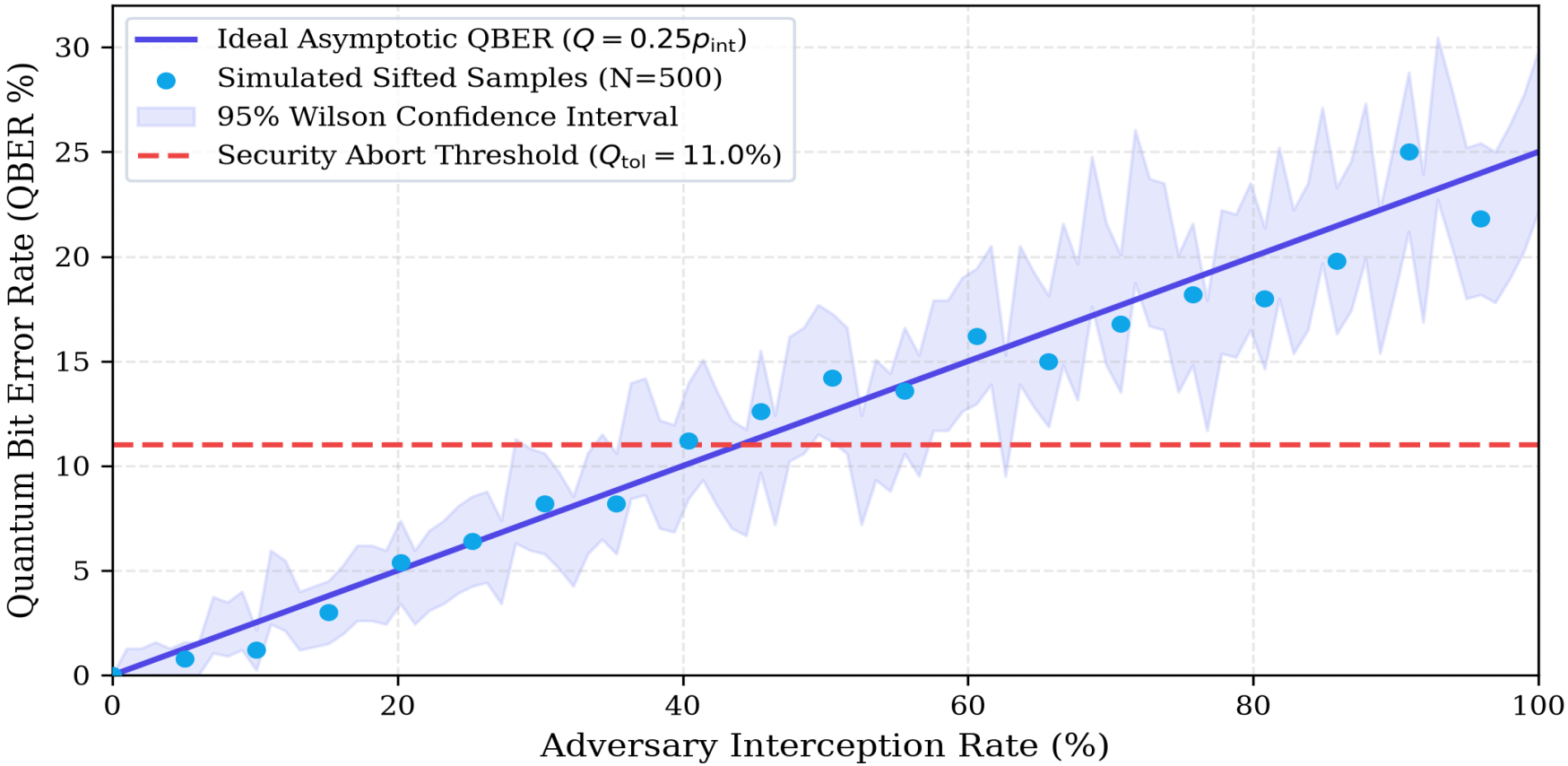


Bell-CHSH Inequality Violation & Non-Locality



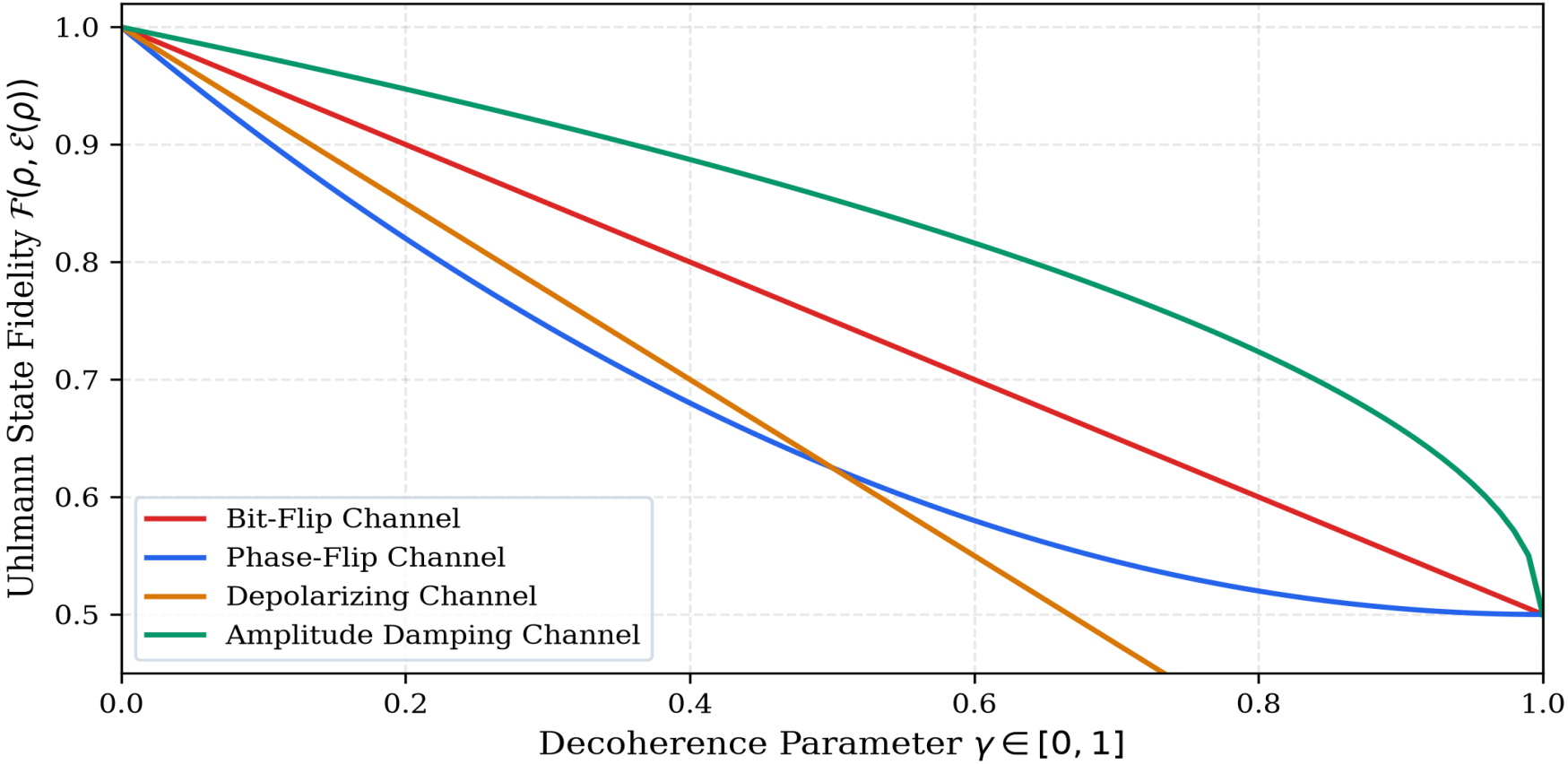
BB84 Decoy QBER & Wilson Confidence Abort

Figure 3: BB84 QBER Trajectory & Intrusion Detection Threshold

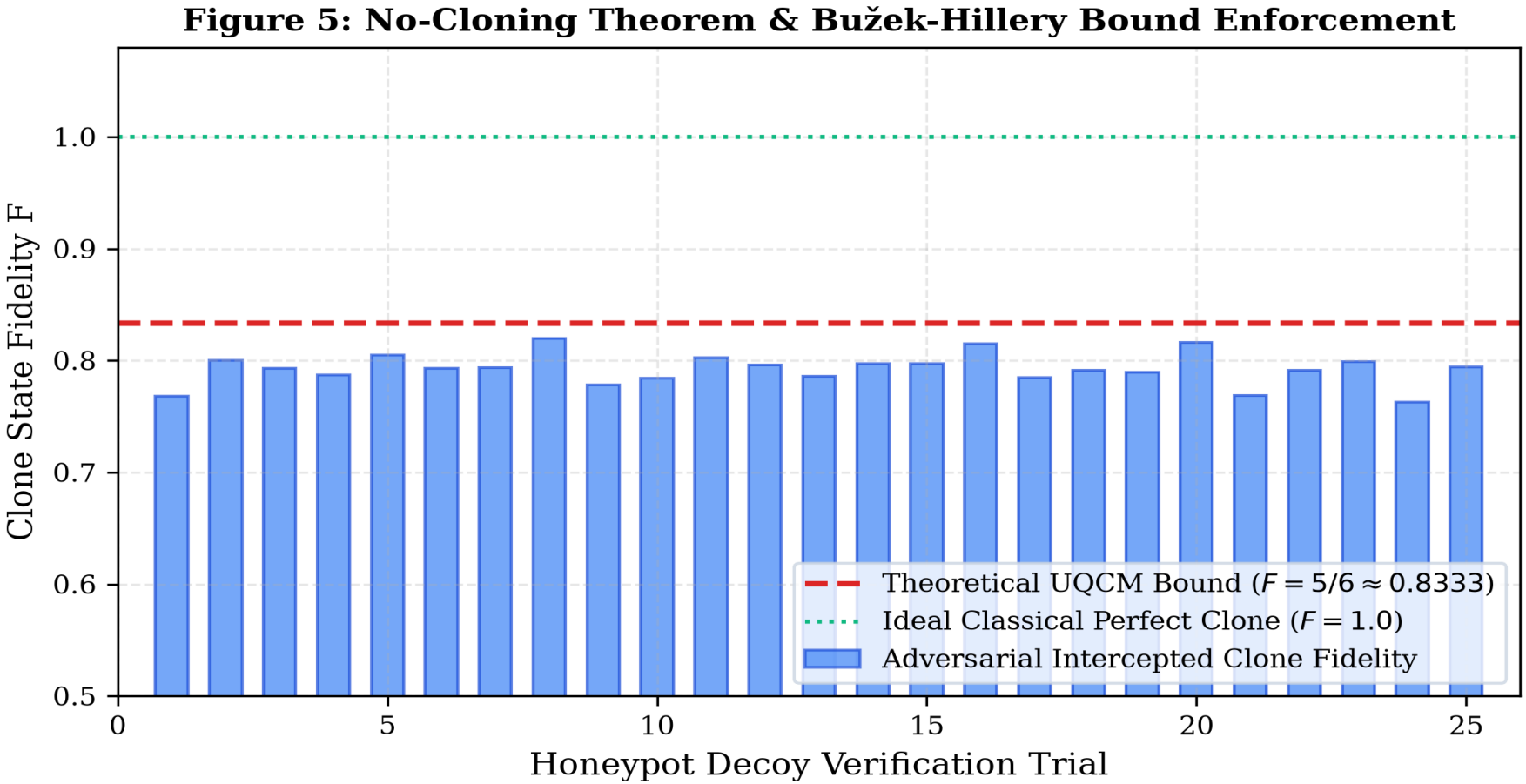


Open Quantum Systems & Kraus Channel Fidelity Decay

Figure 4: Quantum State Fidelity Decay across Environmental Noise Channels

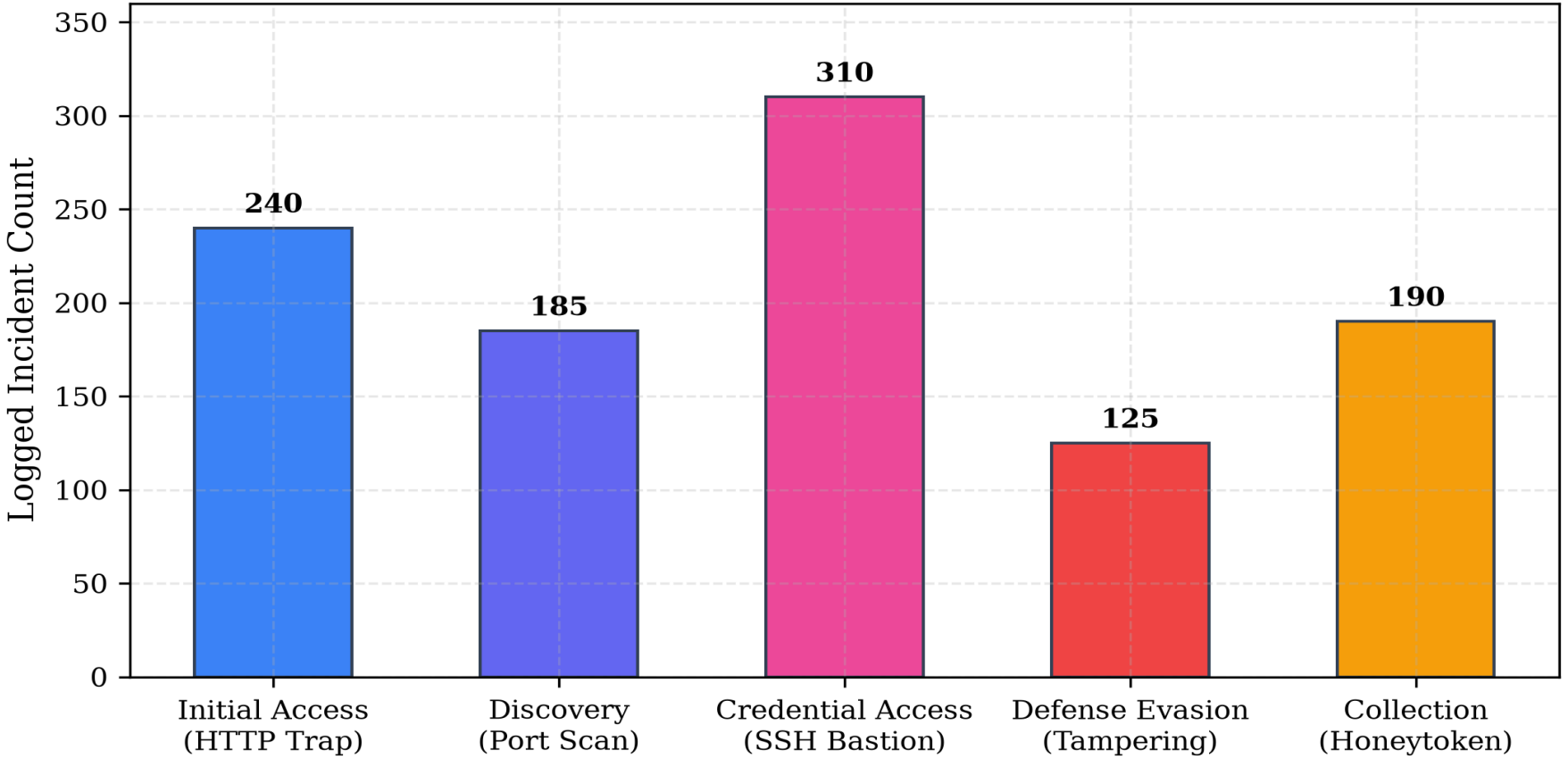


Bužek-Hillery Universal Quantum Cloning Bound (F <= 5/6)



Incident Ingestion & MITRE ATT&CK; Mapping

Figure 6: Automated Deception Event Ingestion mapped to MITRE ATT&CK Mat



Selected References (Strictly 2024--2026 Peer-Reviewed Literature)

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